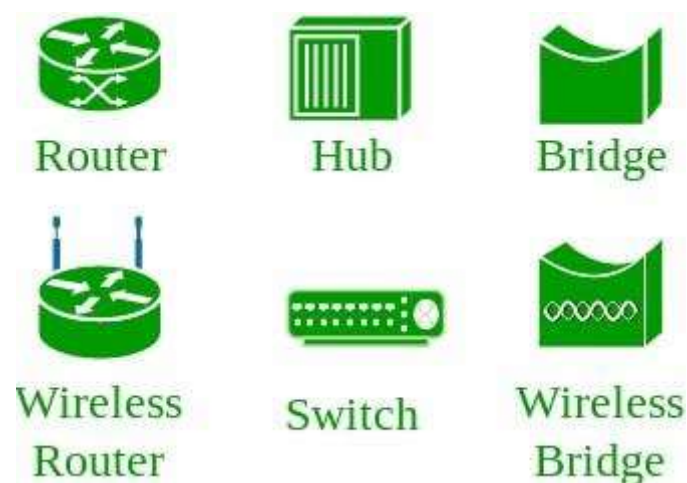


Network Devices

An interconnection of multiple devices, also known as hosts, that are connected using multiple paths for the purpose of sending/receiving data or media. Computer networks can also include multiple devices/mediums which help in the communication between two different devices; these are known as Network devices and include things such as routers, switches, hubs, and bridges.



Various Network devices are:

1. Router

- Connects multiple networks (like home network to the internet).
- Directs data packets to their correct destination.

2. Switch

- Connects devices within a network (like computers in an office).
- Forwards data only to the specific device it is meant for.

3. Hub

- Basic device that connects multiple devices in a network.
- Sends data to all devices (less efficient than a switch).

4. Bridge

- Connects and filters traffic between two networks or segments.
- Helps reduce network traffic.

5. Gateway

- Connects two different types of networks.
- Translates data between different protocols.

6. Access Point (AP)

- Provides wireless connectivity to devices.
- Extends a wired network into a Wi-Fi network.

7. Modem

- Converts digital data from a computer into signals for phone/cable lines and vice versa.
- Provides internet access.

8. Firewall

- Monitors and controls incoming and outgoing network traffic.
- Provides security by blocking unauthorized access.

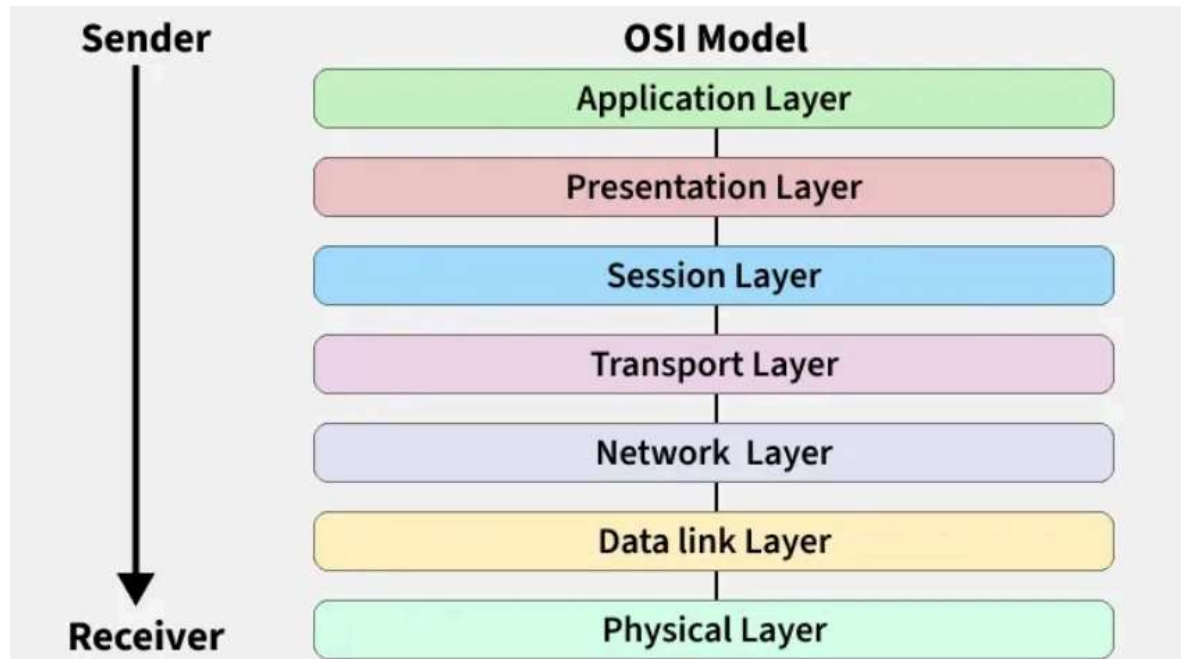
Network Models

In this section foundational frameworks of Computer Networks will be discussed. These frameworks provide a standard set of rules for network communication. We will dive deep into the layered structures of the OSI and TCP/IP models.

- OSI Model
- TCP/IP Model

What is OSI Model? - Layers of OSI Model

The OSI (Open Systems Interconnection) Model is a set of rules that explains how different computer systems communicate over a network. OSI Model was developed by the International Organization for Standardization (ISO). The OSI Model consists of 7 layers and each layer has specific functions and responsibilities. This layered approach makes it easier for different devices and technologies to work together.



OSI Model – Overview of Layers

Layer 1: Physical Layer: The Physical Layer transmits raw data bits over a physical medium using electrical, optical, or radio signals.

- Deals with actual hardware and signals
- Sends data as bits (0s and 1s)
- Examples: Cables, Hub, Repeater
- Focus: *How data moves physically*

Layer 2: Data Link Layer: The Data Link Layer ensures error-free delivery of data frames between two directly connected devices using MAC addresses.

- Ensures error-free data transfer between two devices
- Uses MAC addresses
- Data is called a Frame
- Examples: Switch, Bridge
- Focus: *Safe delivery between neighboring devices*

Layer 3: Network Layer: The Network Layer is responsible for logical addressing and routing packets between different networks.

- Decides which path data should take
- Uses IP addresses
- Data is called a Packet
- Examples: Router
- Focus: *Finding the best route*

Layer 4: Transport Layer: The Transport Layer provides end-to-end communication, segmentation, and reliable or unreliable data delivery using port numbers.

- Provides end-to-end communication
- Uses port numbers
- Data is called a Segment

- **Protocols:** TCP, UDP
- **Focus:** *Complete and reliable delivery*

Layer 5: Session Layer: The Session Layer establishes, manages, and terminates communication sessions between applications.

- **Manages session (connection) between devices**
- **Starts, maintains, and ends communication**
- **Focus:** *Session control*

Layer 6: Presentation Layer: The Presentation Layer translates, encrypts, decrypts, and compresses data into a format understandable by the application layer.

- **Translates data into readable format**
- **Handles encryption, decryption, compression**
- **Focus:** *Data format and security*

Layer 7: Application Layer: The Application Layer provides network services directly to end-user applications.

- Closest to the user
- Provides network services to applications
- Protocols: HTTP, FTP, SMTP, DNS
- Focus: *User interaction with network*

How Data Flows in the OSI Model?

When we transfer information from one device to another, it travels through 7 layers of the OSI model. First data travels down through 7 layers from the sender's end and then climbs back 7 layers on the receiver's end. Data flows through the OSI model in a step-by-step process:

- **Application Layer:** Applications create the data.
- **Presentation Layer:** Data is formatted and encrypted.
- **Session Layer:** Connections are established and managed.
- **Transport Layer:** Data is broken into segments for reliable delivery.
- **Network Layer:** Segments are packaged into packets and routed.
- **Data Link Layer:** Packets are framed and sent to the next device.

- **Physical Layer:** Frames are converted into bits and transmitted physically.

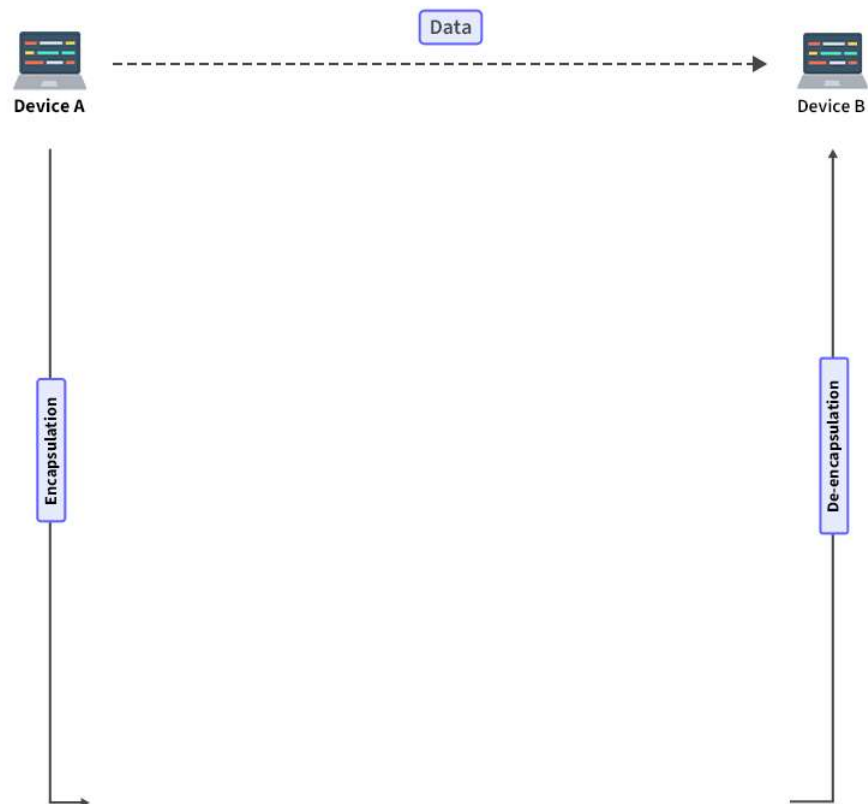
Each layer adds specific information to ensure the data reaches its destination correctly, and these steps are reversed upon arrival.

We can understand how data flows through OSI Model with the help of an example mentioned below. Let us suppose, Person A sends an e-mail to his friend Person B.

- **Step 1: Person A** interacts with e-mail application like Gmail, outlook, etc. Writes his email to send. (This happens at Application Layer).
- **Step 2: At Presentation Layer**, Mail application prepares for data transmission like encrypting data and formatting it for transmission.
- **Step 3: At Session Layer**, there is a connection established between the sender and receiver on the internet.
- **Step 4: At Transport Layer**, Email data is broken into smaller segments. It adds sequence number and error-checking information to maintain the reliability of the information.
- **Step 5: At Network Layer**, addressing of packets is done in order to find the best route for transfer.

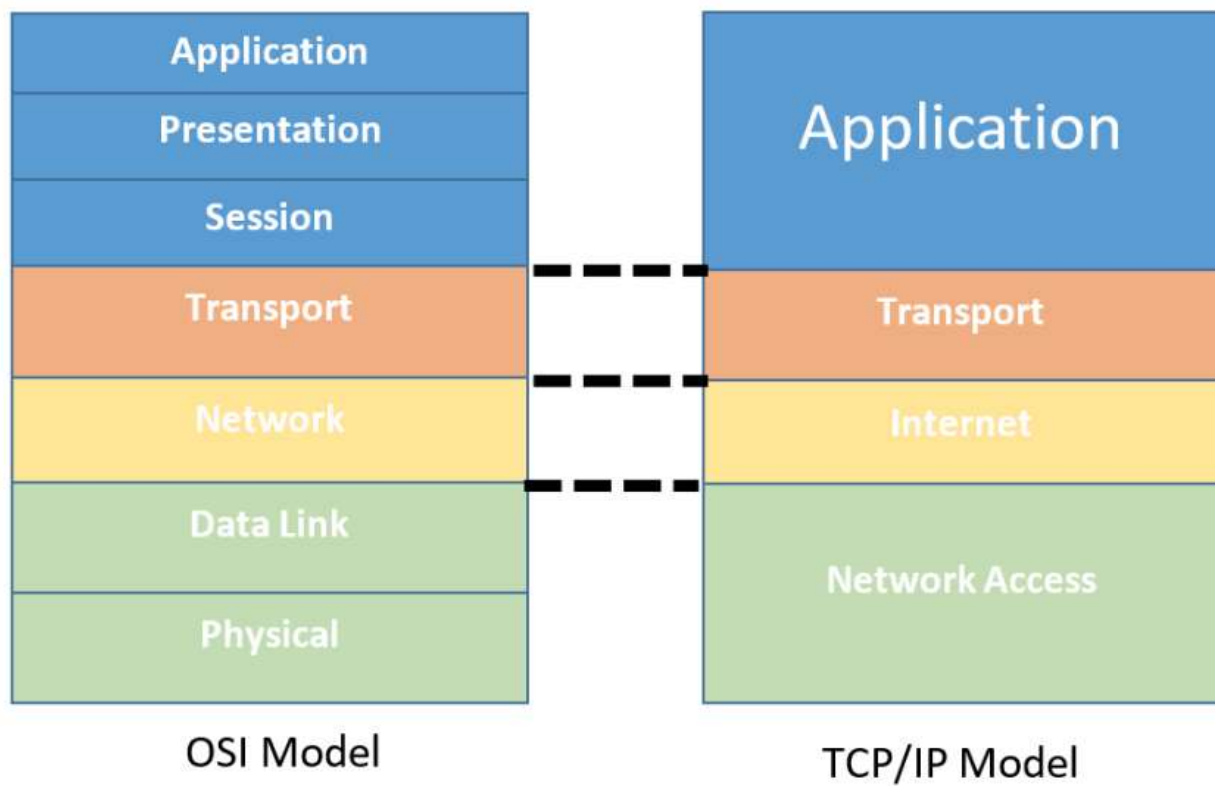
- **Step 6: At Data Link Layer,** data packets are encapsulated into frames, then MAC address is added for local devices and then it checks for error using error detection.
- **Step 7: At Physical Layer,** Frames are transmitted in the form of electrical/ optical signals over a physical network medium like ethernet cable or WiFi.

After the email reaches the receiver i.e. Person B, the process will reverse and decrypt the e-mail content. At last, the email will be shown on Person B email client.



* PH : Presentation Header

Transmission Control Protocol / Internet Protocol (TCP/IP model)



1) What Is the TCP/IP Model?

The **TCP/IP model** (Transmission Control Protocol/Internet Protocol) is a **layered framework used to design and implement network communications**, especially the **Internet** and modern networks. It organizes how data travels from one device to another across networks.

2) Why TCP/IP Is Used Today

TCP/IP is the primary model used in real networks and on the Internet because:

- It was developed along with real protocols that work in the real world, not just theory.
- It is **simpler and practical**, making implementation easier.
- It provides **open standards** widely adopted across devices and systems.
- The model directly maps to the actual suite of protocols used (like TCP, IP, HTTP, DNS).
- **The modern Internet is based on the TCP/IP.**

3) Why Not Use the OSI Model Instead?

The **OSI model** (7-layer) is **mostly theoretical** good for learning and understanding network functions, but not used directly in real implementations.

👉 OSI was created as a **reference model** (conceptual framework), while

👉 **TCP/IP is a real implementation model with working protocols.**

Summary of differences:

- OSI has **7 layers**, TCP/IP typically has **4 or 5 layers**.
- OSI layers are strictly separated; TCP/IP layers may group functions.
- OSI is general; TCP/IP directly maps to real protocols used today.

4) TCP/IP Model Layers

*(Shown here in **5-layer view** commonly used in notes and classes)*

🌐 Layer 5: Application Layer

- **What it is:** The top layer where **user programs interact with network services**.

- **Role:** Provides network services to applications (web browsers, email, FTP, DNS).
- **Functions:** Handles protocols like **HTTP, FTP, SMTP, DNS**.
- **Why:** Combines OSI's **Application, Session, and Presentation** functions into one layer.

Layer 4: Transport Layer

- **What it is:** Provides **end-to-end communication** between hosts.
- **Role:** Ensures the data gets delivered correctly or quickly, based on protocol.
- **Main Protocols:**
 - **TCP** – Reliable, connection-oriented transport.
 - **UDP** – Fast, connectionless transport.
- **Functions:** Segmentation, flow control, error control, sequencing.

Layer 3: Internet Layer

- **What it is:** Responsible for **routing and addressing** data to the right destination.
- **Role:** Adds **IP addresses** and sends data across multiple networks.

- **Key Protocols:** IP (Internet Protocol), ICMP, ARP.
- **Functions:** Routing, logical addressing, packet forwarding.

Layer 2: Data Link Layer

- **What it is:** Handles **node-to-node communication and physical addressing**.
- **Role:** Frames data and prepares it for transmission over physical media.
- **Functions:** Error detection, MAC addressing, access control.
- In 4-layer versions, it is included in the **Network Access** layer.

Layer 1: Physical Layer

- **What it is:** Deals with **actual hardware transmission** (electrical / optical / radio signals).
- **Role:** Sends raw bits over wires, fiber, or air.
- **Devices:** Cables, NICs, hub, modem.
- In some notes, it is separate; in others it is part of the **Network Access** layer.

6) Why Use TCP/IP in Real Networks

Advantages of TCP/IP:

- ✓ Real, working protocols originally designed for the Internet.
- ✓ Flexible, robust and adaptable across many hardware types.
- ✓ Open standards, not controlled by a single vendor or body.
- ✓ Scalable for both small and large networks.

OSI is still important for learning and troubleshooting, but TCP/IP is the actual model used by the Internet and most networking systems today.